

Ex.no:06

10/09/26

Hamming Code - Error detection & Correction.

Aim:-

To write Hamming code for error detection & correction.

Code:-

```
def hamming-encryption(data):
```

```
    k = len(data)
```

```
    r = 0
```

```
    while (2**r) <= (k+r+1):
```

```
        r += 1
```

```
    for
```

```
    n = k + r + 1
```

```
    code = ['0'] * n
```

```
    j = 0
```

```
    for i in range(1, n):
```

```
        if (i & (i-1)) == 0:
```

```
            code[i] = data[j]
```

```
            j += 1
```

```
    for i in range(r):
```

```
        pos = 2**i
```

```
        par = 0
```

```
        for j in range(1, n):
```

```
            if j & pos:
```

```
                par ^= int(code[j])
```

```
    code[pos] = str(par)
```

```
    return (code[1:n], r)
```

```
def hamming code - decryption (data, r):
```

```
    n = len(data)
```

```
    data = ['0'] * list(data)
```

```
    error = 0
```

```
    for i in range(r):
```

```
        pos = 2**i
```

```
        pairs = 0
```

```
        for j in range(1, n+1):
```

```
            if j & pos:
```

```
                pairs += int(data[j])
```

```
            if pairs != 0:
```

```
                error += pos
```

```
    if error != 0:
```

```
        print("Error at position :", error)
```

```
        if (data[error] == '0'):
```

```
            data[error] = '1'
```

```
        else:
```

```
            data[error] = '0'
```

```
    else:
```

```
        print("no error")
```

```
    return ''.join(data[1:])
```

```
data = input("Enter binary data : ")
```

```
encoded, r = hamming code - encryption (data)
```

```
print("Data after encoding from Sender Side :", ''.join
```

```
      (encoded))
```

```
rdata = input("Enter data received by Receiver : ")
```

```
decoded = hamming code - decryption (rdata, r)
```

```
print("Data after decoding :", decoded)
```

Enter binary data = (01) (xlob) rel on

Enter data received by receiver : 0110111

Error at position : 5

Data after decoding: 0110011

Results

Thus, the Hamming code was detected and corrected the error successfully.